🏛️ H.D.R.A. Platform | Hazards & Disasters Risk Assessment

📖 Overview

The **H.D.R.A. (Hazards and Disasters Risk Assessment) Platform** is a next-generation WebGIS and econometric tool designed for the Real Estate and Insurance sectors (PropTech/InsurTech).

It provides automated, highly accurate estimations of how natural disasters (Wildfires, Floods, Earthquakes) impact the present commercial value of a property. By combining live satellite data, a 50-year historical climate/seismic horizon, and advanced econometric modeling, H.D.R.A. transforms raw geospatial data into actionable financial metrics.

> ⚠️ **Repository Notice:** This public repository acts as a showcase of the platform's UI/UX, system architecture, and sample generated reports. The core Python Backend (containing the proprietary Hedonic Pricing Engine and Geospatial parsing algorithms) is strictly closed-source to protect Intellectual Property.

✨ Key Features (The "What")

- 🌍 **Interactive WebGIS Interface:** A dynamic map environment (built with Leaflet.js) allowing users to select any property location and instantly draw an 8-mile analysis buffer zone.
- 📡 **Live Multi-Hazard Scanning:** Integrates seamlessly with global scientific databases to fetch live and 50-year historical data on seismic activity, extreme weather, and active wildfires.
- 🏗️ **Structural Vulnerability Profiling:** Evaluates the unique "DNA" of the building (construction material, age/standards, floor level, roof type) against local hazards.
- 📉 **Actuarial & Hedonic Financial Engine:**
    - Calculates the property's specific **Exposure Factor (EF)**.
    - Computes **SLE** (Single Loss Expectancy) and **ALE** (Annualized Loss Expectancy).
    - Outputs the immediate Hedonic Value Depreciation (Market Drop %) based on the Dynamic Risk Score.

- 📄 **Automated Executive Reporting:** Automatically compiles findings into a professional, multi-page PDF report featuring spiderweb vulnerability charts and hybrid risk-radius maps.

🛠️ Technology Stack & Integrations

Frontend (Publicly Available Here)

- **HTML5, CSS3, Vanilla JavaScript:** Lightweight, lightning-fast, and responsive UI without heavy frameworks.
- **Leaflet.js:** Advanced spatial rendering and interactive map controls.
- **Cinematic UI/UX:** Integrated splash screens, modals, and dynamic data-binding.

Backend & Econometrics (Closed-Source)

- **Python & FastAPI/Uvicorn:** High-performance, asynchronous REST API.
- **Concurrent Execution:** Utilizing `ThreadPoolExecutor` for parallel fetching of multiple external APIs in under 3 seconds.
- **Data Visualization:** `Matplotlib` and `NumPy` for generating complex geospatial hybrid maps and multi-axis radar charts on the fly.
- **Report Generation:** `FPDF2` for dynamic PDF construction.

🛰️ Live Data Providers (Integrated APIs)

- **NASA EONET:** Live tracking of severe active wildfires.
- **Copernicus EU-DEM (via OpenTopoData):** High-resolution digital elevation modeling to calculate terrain slope/ponding risks.
- **EMSC (European-Mediterranean Seismological Centre):** Live and historical seismic fault and magnitude data.
- **Open-Meteo / Copernicus ERA5:** 50-year historical weather scanning (precipitation sum, fire-weather index, wind velocity).

📁 Repository Structure

```
None
├── frontend/
│   ├── index.html           # The main interactive WebGIS Interface
│   └── assets/              # Public assets (Logos, Intro Videos)
├── architecture/
```

```
|   └── flowchart.png         # High-level System Architecture diagram
├── Sample_Reports/           # 🌟 BROWSE THESE TO SEE THE OUTPUT 🌟
|   ├── Report_Total.pdf      # Sample output for Total Multi-Hazard Risk
|   ├── Report_Fire.pdf       # Sample output for Wildfire Risk
|   ├── Report_Flood.pdf      # Sample output for Flood Risk
|   └── Report_Seismic.pdf    # Sample output for Seismic Risk
└── README.md                 # This documentation file
```

📊 Sample Outputs

We highly encourage checking the `Sample_Reports/` directory. These PDFs are completely generated by code in real-time and demonstrate the platform's ability to:

1. Generate comparative **Spiderweb (Radar) Charts** contrasting the specific property's vulnerability against the regional baseline.
2. Render **Hybrid Satellite Maps** visualizing the impact radius of historical disasters relative to the selected coordinates.
3. Output the final **Econometric Valuation** (Value Drop %).

⚖️ License & Copyright

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For live demonstrations, academic inquiries, or investment opportunities, please contact the repository owner directly.